

## B.Tech. Examination-2024

Electronics and Communication Engineering Department  
(Even Semester Regular and Supplementary for ECE & EEN)

Course Title: Digital Electronics and Logic Design

Course Code: ECEUGPC05

Full Marks : 80

Time-3 hrs

Answer Question no. 1 with proper reasoning and any four from the rest  
(Symbols have their usual meaning)

| Q.No   | Questions                                                                                                                                                                            | Marks |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. (a) | Determine the value of x if $(211)_x = (152)_8$ .                                                                                                                                    | 3     |
| (b)    | Design a NAND gate by using 2:1 MUX.                                                                                                                                                 | 3     |
| (c)    | Find the Boolean expression of the following circuit and identify the logic gate in Fig 5.                                                                                           | 3     |
| (d)    | Match the logic gates in Column I with their equivalents in Column II shown in Fig. 1 as shown below.                                                                                | 3     |
| (e)    | In the circuit shown in Fig. 2, X is given by<br>(i) $A \odot B \odot C$ (ii) $A \oplus B \oplus C$ (iii) $A + B + C$ (iv) $ABC$                                                     | 3     |
| (f)    | The logic diagram shown in Fig. 3 is<br>(i) Half-adder (ii) Full-adder (iii) Half-subtractor (iv) Multiplier                                                                         | 3     |
| (g)    | In the circuit shown in Fig. 4, the race around condition<br>(i) does not occur (ii) occur when Clk = 0 (iii) occur when Clk = 0 & A = B = 1<br>(iv) occur when Clk = 0 & A = B = 0. | 2     |

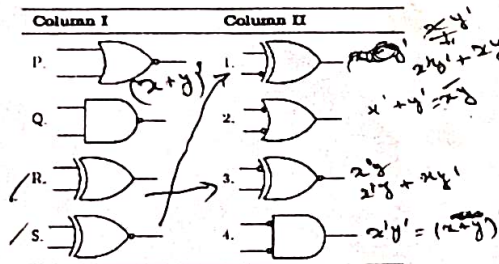


Fig. 1

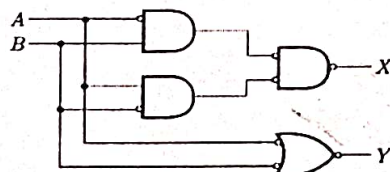


Fig. 3

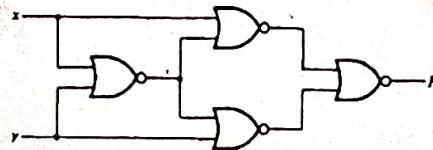


Fig. 5

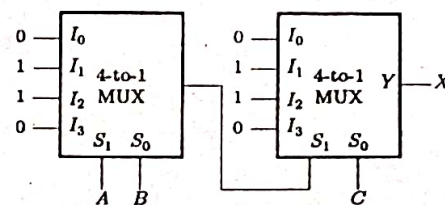


Fig. 2

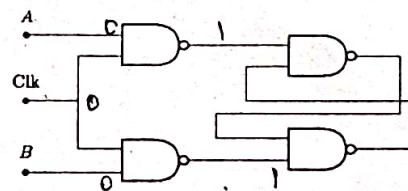


Fig. 4

2. (a) Prove the following Boolean logic identities with algebraic methods only :-  
 (i)  $xy + \bar{x}z + yz = xy + \bar{x}z$  (ii)  $x \oplus y \oplus z = x \odot y \odot z$  (iii)  $\overline{A+B} = \bar{A} \cdot \bar{B}$  3×4
- (b) Show that NOR gate is an universal logic element. 3
3. (a) Explain the operation of 4-bit binary parallel adder/subtractor. 4
- (b) Design a BCD-to-excess-3 code converter. 6
- (c) Implement a Full-Subtractor circuit with a  $3 \times 8$  decoder and two OR gates. 5
4. (a) Design a 3-bit Comparator Circuit by using basic gates. 5
- (b) Design a look ahead carry generator for a 4-bit serial adder. 5
- (c) Implement a Full Adder circuit with a decoder and OR gate. 5

5. (a) Simplify the following function using K-Map considering the don't care condition 8  
 $F(x, y, z, w) = \sum(0, 1, 2, 3, 7, 8, 10)$ ,  $D(x, y, z, w) = \sum(5, 6, 11, 15)$
- (b) Obtain the simplified expression in product of sums form of the following logic function. 7  
 $F(w, x, y, z) = \prod(1, 3, 5, 7, 13, 15)$
6. (a) Implement  $8 \times 1$  MUX with two  $4 \times 1$  MUX and OR gate. 7
- (b) Implement the following function with a multiplexer: 8  
 $F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 11, 15)$
7. (a) Draw and explain the operation of mod-8 ripple counter with output timing waveforms. 7
- (b) Show that maximum clock frequency of the counter is  $f_{max} = \frac{1}{Nt_{pd} + T_s}$  8
8. (a) Design a 4 bit asynchronous up/Down counter by using T-flip-flop. 6
- (b) Describe the design of a 4-bit Ring counter. 4
- (c) Draw a 4 bit universal shift register with parallel load and explain its operation. 5
9. (a) Draw a logic diagram of a clocked master slave SR flip-flop and explain its operation. 6
- (b) Realize JK flip-flop from SR flip-flop and verify its truth table. 3
- (c) Convert JK flip-flop into D flip-flop. 8
10. (a) With suitable logic diagram explain the operation of mod-4 ring counter. 7
- (b) Explain the operation of 4-bit shift register with timing waveform. 4+4
11. (a) Design NAND, NOR gate by using CMOS and explain its operation. 2
- (b) Design 2-input NOR gate by using MOS. 5
- (c) Design a basic TTL NAND gate and explain its operation.

0 x  
 1 0  
 0 1  
 x 0



**Aliah University**  
**Department of Electronics & Communication Engineering**  
**(B.Tech, 4<sup>th</sup> Sem, 2024)**

Paper Name: Computer Organization

Paper Code: CSEUGOE02

Time: 3 hours

F.M: 80

**Group-A**  
**(Multiple Choice Type Questions)**

Answer the following questions.

1X10=10

1.
  - i. Write down the functions of a Control Unit?
  - ✓ ii. Describe the term "Instruction Cycle"?
  - iii. What do you mean by TLB?
  - iv. What does SIMD represent in computer organization?
  - ✓ v. Write the function of the *Program Counter*.
  - ✓ vi. Write the value of (100101) after the *Arithmetic Shift Right* of the *Register content* twice.
  - ✓ vii. Represent the no. (-23) in 2's complement (8-bit format).
  - viii. What does '*locality of Reference*' mean?
  - ix. *Stack* uses which type of addressing modes?
  - x. What do you mean by the term *Interrupt*?

**Group-B**  
**(Short Answer Type Questions)**

5X6=30

Answer any six questions.

2. Explain Auto-decrement Addressing with proper examples. If the value of the *Register* is 400, then find the effective address and accumulator content for *Direct, Indirect, Immediate & Register Indirect* mode. (1+4)
3. Apply Booth's algorithm to multiply the two numbers (-7) and (4). Assume the multiplicand and multiplier to be 4 bit each. (5)

|                  |     |
|------------------|-----|
| Address = 300    | 201 |
| Next Instruction | 202 |
|                  |     |
| 350              | 299 |
| 600              | 300 |
|                  |     |
| 700              | 399 |
| 500              | 400 |
|                  |     |
| 100              | 600 |

- ✓ 4. A cache has a 1MB capacity with Block size 64B and 4-way set associative. The CPU generates a 32-bit address for accessing data in the memory. How many bits are required to represent the Block offset, Set no. and Tag field? (5)
- ✓ 5. Explain the difference among two address and one address instruction for the equation  $X = (A * B)$ . Represent 37.5 in IEEE-754 single precision format. (2+3)  
1.1875
- ✓ 6. Find the addition of (+1.5) & (-0.125) using 8-bit representation. Explain the Write-through and Write-back mechanism in memory? (3+2)



7. Discuss the Memory Hierarchy with its characteristics. (5)
8. Explain Page and Frame both are same or not. Given- effective access time = 200 ns, Main memory access time = 100 ns, TLB Hit ratio = 70%, Find the Miss ratio and TLB access time. (2+3)

**Group-C**  
(Long Answer Type Questions)

10x4=40

Answer any four questions.

9. Write the difference between memory mapped I/O and I/O mapped I/O. Draw the space-time diagram for 4stage pipeline and non-pipelined architecture to complete 6 tasks and find the *efficiency*. In a pipeline architecture, the time delay for 4 stages  $\tau_1=120\text{ns}$ ,  $\tau_2=140\text{ns}$ ,  $\tau_3=1600\text{ns}$ , and  $\tau_4=190\text{ns}$ , and their interface latch has a delay of 20ns then, find the Speed Up ratio. (2+4+4)
10. Explain software based priority interrupt with diagram. Discuss the concept of a reservation table for linear and non-linear pipelining architecture with diagram. Find Permissible latency and Collision Vector for the following reservation table. (3+3+4)

|    | 1              | 2              | 3              | 4              | 5              | 6              | 7              | 8              | 9              |
|----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| S1 | X <sub>1</sub> |                |                |                |                | X <sub>2</sub> |                | X <sub>3</sub> |                |
| S2 |                | X <sub>1</sub> |                | X <sub>2</sub> |                |                | X <sub>3</sub> |                | X <sub>4</sub> |
| S3 |                |                | X <sub>1</sub> |                | X <sub>2</sub> |                |                |                | X <sub>3</sub> |

11. Explain different *Cache Memory* mapping techniques in details. 10

12. Define the term Access time, Seek time, and Latency in case of accessing Secondary Memory. Consider a fully associative cache with a total of 5 *cache blocks* (0-4). The main memory block requests are as follows: 4,3,25,8,19,6,25,8,16,35,45,22,8,3,16,25,7. Assume initially the cache is empty. Finds the blocks currently present in Cache Memory using FIFO and LRU technique. (3+7)

13. Write short notes in any *two* of the following: 5+5

- i. Virtual memory
- ii. Associative Memory
- iii. RAID

X



**B.Tech Examination-2024**  
**Electronics and Communication Engineering**  
**(Even Semester Regular and Supplementary)**

**Course Name: Electromagnetic Engineering, Course Code: ECEUGPC06**

**Full Marks: 80**

**Time: 3.00 Hrs**

- Answer Q.1 and six questions from rest of the questions.
- Answer all parts of a question in same place.
- Symbols have their usual meaning.

**1. Choosing the correct alternative from the given options answer any ten questions.**

- (a) Which of the following expressions results into zero?  
 (i)  $\nabla(\nabla \cdot \mathbf{A})$  (ii)  $\nabla(\nabla \times \mathbf{A})$  (iii)  $\nabla(\nabla \cdot \mathbf{V})$  (iv)  $\nabla \times (\nabla \times \mathbf{A})$  2
- (b) The force between two electric charges varies with the distance between them. The variation is \_\_\_\_\_ to the square of that distance.  
 (i) equal (ii) greater than (iii) directly proportional (iv) inversely proportional 2
- (c) Conduction current density is given as  
 (i)  $\mathbf{j} = \frac{1}{\sigma E}$  (ii)  $\mathbf{j} = \sigma E$  (iii)  $\mathbf{j} = \frac{1}{\rho E}$  (iv)  $\mathbf{j} = \rho E$  2
- (d) Equation of continuity is based on  
 (i) conservation of charge (ii) conservation of momentum (iii) conservation of angular momentum (iv) None of this 2
- (e) Poynting theorem represents  
 (i) conservation of charge (ii) conservation of momentum (iii) conservation of energy (iv) none of this 2
- (f) According to Maxwell's equation in free space  $\nabla \cdot \mathbf{E} = ?$   
 (i)  $\rho$  (ii)  $\frac{\rho}{\epsilon_0}$  (iii) 0 (iv)  $\frac{\epsilon_0}{\rho}$  2
- (g) Maxwell observed and corrected a discrepancy in  
 (i) Ampere's Law (ii) Faraday's Law (iii) Gauss's Law for Electromagnetic (iv) Snell's law 2
- (h) Gauss's law for magnetism is  
 (i)  $\nabla \cdot \mathbf{B} = 0$  (ii)  $\nabla \times \mathbf{H} = \mathbf{D} + \mathbf{J}$  (iii)  $\nabla \times \mathbf{E} = -\mathbf{B}$  (iv)  $\nabla \cdot \mathbf{D} = 0$  2
- (i) The wave velocity in non-conducting medium is  
 (a)  $\frac{1}{\sqrt{\mu\epsilon}}$  (b)  $\sqrt{\frac{\mu}{\epsilon}}$  (c)  $\frac{\mu_0}{\epsilon_0}$  (d)  $\frac{1}{\sqrt{\mu_0\epsilon_0}}$  2
- (j) A transmission line is said to be distortion less line when  
 (i)  $\frac{R}{L} = \frac{G}{C}$  (ii)  $\frac{R}{G} = \frac{C}{L}$  (iii)  $RG = \frac{L}{C}$  (iv)  $\frac{R}{G} = LC$  2
- (k) The magnitude of the induced e.m.f in a conductor depends on the  
 (i) the flux density of the magnetic field (ii) amount of flux cut (iii) amount of flux-linkages (iv) rate of change flux-linkages 2
- (l) The direction of propagation of electromagnetic wave is obtained from  
 (i)  $\mathbf{E} \times \mathbf{H}$  (ii)  $\mathbf{E} - \mathbf{H}$  (iii)  $\mathbf{E}$  (iv)  $\frac{\mathbf{E}}{\mathbf{H}}$  2
- (a) State Divergence theorem and Stoke's theorem. 4
- (b) Show that the vector  $\mathbf{H} = (y^2 - z^2 + 3yz - 2x) \mathbf{a}_x + (3xz + 2xy) \mathbf{a}_y + (3xy - 2xz + 2z) \mathbf{a}_z$  is both irrotational and solenoidal. 3
- (c) Using the cylindrical coordinate system find the area of a curved surface on the right of a circular cylinder having radius = 3m, height = 6m and  $30^\circ \leq \phi \leq 120^\circ$  3
3. (a) What do you mean by electric field intensity at a point? 2
- (b) Define electric flux and flux density. 2

$$\oint (\nabla \times \vec{A}) \cdot d\vec{s} = \oint \vec{A} \cdot d\vec{s}$$

$$\oint (\nabla \times \vec{A}) \cdot d\vec{s} = \oint \vec{A} \cdot d\vec{s}$$



- (c) Find the force on a point charge  $q$  located at  $(0,0,h)$  m due to charge of surface charge density  $\rho_s$  C/m<sup>2</sup> uniformly distributed over the circular disc  $r \leq a$ ,  $z = 0$  m. 6
4. (a) State and prove the Gauss's Law.  $\oint \vec{D} \cdot d\vec{s} = Q$  5  
 (b) Write Poisson's and Laplace's equation. 2  
 (c) Let  $\vec{D} = 2y^2z^2\vec{a}_x + 3xy^2z^2\vec{a}_y + 2xyz\vec{a}_z$  pC/m<sup>2</sup> in free space. Find (i) the total electrical flux passing through the surface  $x = 2, 0 \leq y \leq 2, 0 \leq z \leq 2$  in a direction away from the origin, (ii) the total charge contained in an incremental sphere of a radius  $1 \mu\text{m}$  centered at  $P(2,2,2)$ . 3
5. (a) Derive the equation of continuity  $\frac{\partial \rho_v}{\partial t} + \nabla \cdot \vec{j} = 0$  3  
 (b) Prove that the normal component of electric flux density is discontinuous across the boundary in presence of surface charge density between two perfect dielectric medium. 4  
 (c) A boundary exists at  $z = 0$  between two dielectrics  $\epsilon_{r1} = 2.5$  region  $z < 0$ , and  $\epsilon_{r2} = 4$  region  $z > 0$ . The field in region  $\epsilon_{r1}$  is  $\vec{E}_1 = -30\vec{a}_x + 50\vec{a}_y + 70\vec{a}_z$  V/m. Find (i) normal component of  $\vec{E}_1$  (ii) tangential component of  $\vec{E}_1$  and (iii) normal component of  $\vec{D}_2$ . 3
6. (a) State Ampere's circuital law. 2  
 (b) Find the magnetic field intensity at a point  $(r, \phi, z)$  due to an infinitely long straight filament carrying a current  $I$  in the  $+z$  direction. 5  
 (c) The flux density at a point distance  $r$  from a long filamentary conductor carrying a current  $I$  in  $\vec{a}_z$  direction is given by  $\vec{B} = \frac{\mu_0 I}{2\pi r} \vec{a}_\phi$ . Calculate the flux crossing the portion of plane  $\phi = \frac{\pi}{4}$  defined by  $0.01 < r < 0.05$  m and  $0 < z < 2$  m for a current of 2 A. 3
7. (a) State Faraday's law of electromagnetic induction. 2  
 (b) Prove that  $\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$ . The symbols have usual meaning. 4  
 (c) Write four Maxwell's equation in integral and differential form and state their significance. 4
8. (a) What are the properties of electromagnetic wave? 2  
 (b) Derive the wave equation in terms of electric field and magnetic field from Maxwell's equation for free space. 5  
 (c) An electric field is represented by  $\vec{E}_e = 10 \cos(6\pi \times 10^8 t - \frac{\pi}{4} x) \vec{a}_y$  is propagating through a lossless medium having  $\mu_r = 1$  and  $\epsilon_r = 78$  at a frequency of 300 MHz. Find  $\beta$ ,  $\eta$  and the corresponding magnetic field  $\vec{H}$ . 3
9. (a) State and prove Poynting Theorem. 5  
 (b) Derive the expression for reflection and transmission coefficients when uniform plane wave incidents normally on the interface between two perfect dielectric medium and also find their relation. 5
10. (a) Draw the equivalent circuit of two wire transmission line. 2  
 (b) Starting from the equivalent circuit derive the transmission line equations for  $V$  and  $I$ , in terms of source parameter. 5  
 (c) Find the expression of propagation constant, phase velocity and characteristics impedance of a lossless transmission in terms of its primary parameters. 3
11. (a) Find the expression for the input impedance  $Z_{in}$  of a short-circuited, an open-circuited and a matched lossless transmission line using standard expression of input impedance of transmission line.  $V^- = \text{reflected}, V^+ = \text{incident}.$  3  
 (b) Define the term VSWR and reflection coefficient for transmission line and write the relation between them. 4  
 (c) A 5 m long lossless transmission line having phase constant equal to  $\pi$  is terminated by a load  $Z_L = 50 + j10 \Omega$ . The values of inductance and capacitance are given to be 200 nH/m and 50 nF/m, respectively. Determine the input impedance of the line at the source end. 3



- (c) Find the force on a point charge  $q$  located at  $(0,0,h)$  m due to charge of surface charge density  $\rho_s$  C/m<sup>2</sup> uniformly distributed over the circular disc  $r \leq a$ ,  $z = 0$  m. 6
4. (a) State and prove the Gauss's Law.  $\oint \vec{D} \cdot d\vec{s} = Q$  5  
 (b) Write Poisson's and Laplace's equation. 2  
 (c) Let  $\vec{D} = 2y^2z^2\vec{a}_x + 3xy^2z^2\vec{a}_y + 2xyza_z$  pC/m<sup>2</sup> in free space. Find (i) the total electrical flux passing through the surface  $x = 2, 0 \leq y \leq 2, 0 \leq z \leq 2$  in a direction away from the origin, (ii) the total charge contained in an incremental sphere of a radius 1  $\mu$ m centered at  $P(2,2,2)$ . 3
5. (a) Derive the equation of continuity  $\frac{\partial \rho_v}{\partial t} + \nabla \cdot \vec{j} = 0$  3  
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 (c) A boundary exists at  $z = 0$  between two dielectrics  $\epsilon_{r1} = 2.5$  region  $z < 0$ , and  $\epsilon_{r2} = 4$  region  $z > 0$ . The field in region  $\epsilon_{r1}$  is  $\vec{E}_1 = -30\vec{a}_x + 50\vec{a}_y + 70\vec{a}_z$  V/m. Find (i) normal component of  $\vec{E}_1$  (ii) tangential component of  $\vec{E}_1$  and (iii) normal component of  $\vec{D}_2$ . 3
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 (b) Find the magnetic field intensity at a point  $(r, \phi, z)$  due to an infinitely long straight filament carrying a current  $I$  in the  $+z$  direction. 5  
 (c) The flux density at a point distance  $r$  from a long filamentary conductor carrying a current  $I$  in  $\vec{a}_z$  direction is given by  $\vec{B} = \frac{\mu_0 I}{2\pi r} \vec{a}_\phi$ . Calculate the flux crossing the portion of plane  $\phi = \frac{\pi}{4}$  defined by  $0.01 < r < 0.05$  m and  $0 < z < 2$  m for a current of 2A. 3
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8. (a) What are the properties of electromagnetic wave? 2  
 (b) Derive the wave equation in terms of electric field and magnetic field from Maxwell's equation for free space. 5  
 (c) An electric field is represented by  $\vec{E}_y = 10 \cos(6\pi \times 10^8 t - \beta x) \vec{a}_y$  is propagating through a lossless medium having  $\mu_r = 1$  and  $\epsilon_r = 78$  at a frequency of 300 MHz. Find  $\beta$ ,  $\eta$  and the corresponding magnetic field  $\vec{H}$ . 3
9. (a) State and prove Poynting Theorem. 5  
 (b) Derive the expression for reflection and transmission coefficients when uniform plane wave incidents normally on the interface between two perfect dielectric medium and also find their relation. 5
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 (c) A 5 m long lossless transmission line having phase constant equal to  $\pi$  is terminated by a load  $Z_L = 50 + j10 \Omega$ . The values of inductance and capacitance are given to be 200 nH/m and 50 nF/m, respectively. Determine the input impedance of the line at the source end. 3

**B.Tech.Examination-2024**  
**Dept. of Electronics and Communication Engineering**  
**(Even Semester Regular & Supplementary)**  
**Course Title: Electronic and Electrical Measurement**  
**Course Code: ECEUGPC07**

Full Marks: 80

Time:3.00Hrs

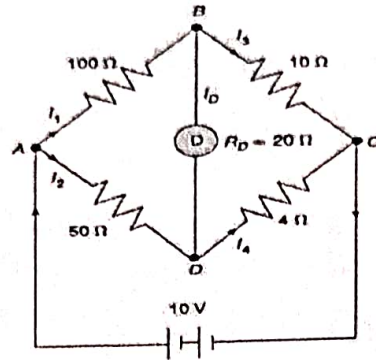
- Answer question no.1 and any four questions from the rest.
- Answer all parts of a question in same place.
- Figures on the right hand side margin indicate full marks.
- Symbols have their usual meaning

| Q. No. | Questions                                                                                                                                                                                                                                                                                                                      | M  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1      | (a) How to increase the voltmeter range and ammeter range in the measurement system?                                                                                                                                                                                                                                           | 2  |
|        | (b) Differentiate between absolute and secondary instruments.                                                                                                                                                                                                                                                                  | 2  |
|        | (c) The measurand value of a resistance is $10.25\Omega$ , where as its value is $10.2\Omega$ . Determine the absolute error of the measurement.                                                                                                                                                                               | 2  |
|        | (d) Explain how eddy current damping is achieved in a Permanent Magnet Moving Coil (PMMC) instrument.                                                                                                                                                                                                                          | 2  |
|        | (e) What are lissajous patterns?                                                                                                                                                                                                                                                                                               | 2  |
|        | (f) Why the lower end of scale of an MI instrument is congested?                                                                                                                                                                                                                                                               | -  |
|        | (g) What is aquadog and swamping resistance?                                                                                                                                                                                                                                                                                   | 2  |
|        | (h) Explain the sensitivity and linearity is two conflicting requirements in a resistance potentiometer.                                                                                                                                                                                                                       | 2  |
| 2      | (a) What are the different types of errors in a measuring instrument? Describe their source briefly.                                                                                                                                                                                                                           | 6  |
|        | (b) The following 10 observations were recorded when measuring a voltage:<br>41.7, 42.0, 41.8, 42.0, 42.1, 41.9, 42.0, 41.9, 42.5, 41.8<br>Find the following parameters:<br>i) Arithmetic Mean<br>ii) Average deviation<br>iii) Standard deviation<br>iv) Variance<br>v) Probable error of one reading.                       | 10 |
| 3      | (a) Describe the generalized measurement system with an appropriate block diagram and explain the loading effect in measurement systems.                                                                                                                                                                                       | 10 |
|        | (b) Explain the following terms associated with measuring instruments preferably with examples: (a) Accuracy, (b) Precision, (c) Reproducibility (d) Linearity (e) Fidelity and (f) Drift                                                                                                                                      | 6  |
| 4      | (a) Sketch the curves showing deflection versus time for analog indicating instrument for under damping, critical damping and over damping.                                                                                                                                                                                    | 3  |
|        | (b) Derive the equation for deflection of a PMMC instrument if the instrument is spring controlled.                                                                                                                                                                                                                            | 7  |
|        | (c) The coil of a PMMC instrument has 60 turns, on a former that is 18 mm wide, the effective length of the conductor being 25 mm. It moves in a uniform field of flux density 0.5 Tesla. The control spring constant is $1.5 \times 10^{-6}$ Nm/degree. Calculate the current required to produce a deflection of 100 degree. | 6  |



- 5 (a) Explain the principle of working of a Kelvin's double bridge for measurement of unknown low resistance. Explain how the effects of contact resistance and resistance of leads are eliminated. 10

(b)



Find the current through the galvanometer that is connected across points B and D having resistance of  $20\Omega$  and a potential difference of  $10V$  across points A and C.

- 6 (a) Explain with the help of phasor diagram, how unknown inductance can be measured using Maxwell's inductance-capacitance bridge. 10

(b) The four arms of a bridge are connected as follows: 6

Arm AB: A capacitor  $C_1$  with an equivalent series resistance  $r_1$

Arm BC: A non inductive resistance  $R_3$

Arm CD: A non inductive resistance  $R_4$

Arm DA: A capacitor  $C_2$  with an equivalent series resistance  $r_2$  in series with a resistance  $R_2$ .

A supply of  $500\text{ Hz}$  is given between terminals A and C and the detector is connected between nodes B and D. At balance,  $R_2 = 5\Omega$ ,  $R_3 = 1000\Omega$ ,  $R_4 = 3000\Omega$ ,  $C_2 = 0.3\mu\text{F}$  and  $r_2 = 0.25\Omega$ . Calculate the values of  $C_1$  and  $r_1$ , and dissipation factor of the capacitor.

- 7 (a) What are the merits of the digital systems over analog system? 3  
(b) How many types of digital voltmeters are there? Explain any one of them briefly. 7  
(c) With the help of a functional block diagram, describe the principle of operation of a digital frequency meter. 6

- 8 (a) Draw the block diagram of a CRO and explain the different components. 6  
(b) Derive an expression for the vertical deflection on the screen of a cathode ray tube in terms of length of plates, separation distance, accelerating voltage and distance of screen from the origin. 10

- 9 (a) Derive the expression of output of the resistance potentiometer when connected to a meter 10  
(b) Total resistance of the potentiometer is  $10K\Omega$  and the length of the potentiometer resistance is  $4\text{mm}$ . the power rating is  $20\text{mW}$ . Determine the output voltage when the displacement of the wiper is  $1.2\text{ mm}$ . assuming the maximum permissible excitation voltage. 6

$$e_o = \frac{2\pi}{2\pi} \cdot e_i$$

Marks: 80

Time: 3hrs

A. Answer the following question by selecting the correct options.

1×30=30

1. The percentage of human genome which encodes proteins is approximately  
A. Less than 2%  
B. 5%  
C. 25%  
D. 99%
2. On the Origin of Species was written by \_\_\_\_  
A. Charles Darwin  
B. Ludmila Kuprianova  
C. Mikhail  
D. Fedonkin
3. The EMP pathway in eukaryotes usually takes place in  
A. nucleus  
B. lysosome  
C. Golgi apparatus  
D. cytosol
4. Plasma membrane is  
A. Permeable  
B. Selectively permeable  
C. Impermeable  
D. Semi-permeable
5. Where is DNA present in the eukaryotic cells?  
A. Inside the nucleus  
B. With other cellular contents  
C. Inside the ribosomes  
D. Not present
6. Which of the following is the function of lysosomes?  
A. Autophagy  
B. Autolysis  
C. Digestion  
D. All of the above
7. Which of the following immunity is present from our birth?  
A. Innate Immunity  
B. Active immunity  
C. Passive immunity  
D. Acquired immunity

8. Which of the following cells is involved in humoral immunity?  
A. T-cells  
B. B-cells  
C. Mast cells  
D. Both T and B cells

9. Name the RNA molecules which is used to carry genetic information copied from DNA?

- A. tRNA  
B. mRNA  
C. rRNA  
D. snRNA

10. Glycolysis is the conversion of

- A. Fructose into phosphoenolpyruvate  
B. Fructose into pyruvate  
C. Glucose into phosphoenolpyruvate  
D. Glucose into pyruvate

11. The jawless vertebrate is

- A. Crocodile  
B. zoris  
C. *Hyla*  
D. *Petromyzon*

12. The genetic materials of Viruses can be \_\_\_\_

- A. ssDNA  
B. ssRNA  
C. dsDNA  
D. All of the above

13. Which of these is a characteristic of prokaryotic cells?

- A. Absence of cell organelles  
B. Absence of nucleus  
C. Presence of 70S ribosomes  
D. All of these

14. Which of the following is a flightless bird?

- A. Pigeon  
B. vulture  
C. Parrot  
D. ostrich

15. Book lungs are respiratory organs in

- A. Insects  
B. Arachnids  
C. Molluscs  
D. Echinoderms

16. This is the largest phylum of Animal on the earth.

- A. Mollusca  
B. Amphibia  
C. Arthropoda  
D. Aves

17. Which of the following nitrogenous base is not present in DNA?

- A. Thymine  
B. Adenine  
C. Guanine  
D. Uracil

18. Who is known as the "Father of Genetics"?

- A. Morgan  
B. Mendel  
C. Watson  
D. Bateson



19. Which substrate is used in the last step of glycolysis?  
 A. dihydroxyacetone 3-phosphate B. Pyruvate  
 C. Phosphoenolpyruvate D. 1, 3-bisphosphoglycerate

20. Anthracene is present in  
 A. DNA B. tRNA  
 C. rRNA D. mRNA

21. From which structure is a mesosome derived from?  
 A. Plasmid B. Cell-wall  
 C. Ribosome D. Cell membrane

22. The percentage of human genome which encodes proteins is approximately  
 A. Less than 2% B. 5%  
 C. 25% D. 99%

23. The scaleless vertebrate is  
 A. Snake B. Rohu  
 C. Shark D. rat

24. Phylum of *Taenia Solium* is  
 A. Aschelminthes B. Annelids  
 C. plathelminths D. Mollusca

25. Sickle cell anemia is caused  
 A. When valine is replaced by glutamic acid in beta polypeptide chain  
 B. When glutamic acid is replaced by valine in beta polypeptide chain  
 C. When glutamic acid is replaced by valine in alpha polypeptide chain  
 D. When valine is replaced by glutamic acid in alpha polypeptide chain

26. In prokaryotes, the hair-like outgrowths which attach to the surface of other bacterial cells are  
 A. Flagella  
 B. Pili  
 C. Capsule  
 D. Plasmids

27. Phylum of *Taenia Solium* is  
 A. Aschelminthes  
 B. Annelids  
 C. plathelminths  
 D. mollusca

28. The jawless vertebrate is  
 A. crocodile  
 B. zonis  
 C. *Hydra*  
 D. *Petromyzon*

29. Which of the following features are common among prokaryotes and eukaryotes?  
 A. Coxi apparatus  
 B. Flagella  
 C. Endoplasmic reticulum  
 D. All of the above

30. In a fetus, where are lymphocytes produced?  
 A. In the spleen  
 B. In the bone marrow  
 C. In the liver  
 D. In the heart

- B. Answer any 5 questions from the following.  
 1. What is Codon?  
 2. Write down the two characteristic features of Phylum Anthropoda.  
 3. What is Coclon? Where is it found?  
 4. What is Nephridia? Where it is found?  
 5. Mention two difference between Adaptive Immunity and Innate Immunity.  
 6. What is chromosome?  
 7. What is tRNA? Mention its one function.

2x5=10

- C. Answer any 4 questions from the following.

5x4=20

1. What is Coclon? Write down the characteristics features Phylum Porifera.  
 2. Short note on B-cell and T-cell. *Plasma Cell, Memory B cell*  
 3. Write down the structure of Placid and mention its function. *affer*  
 4. Describe the Operin and Haldane theory on the chemical basis of origin of life.  
 5. What is notchord? Write down the characteristics features of class Reptilia.  
 6. What is immunity? Mention the different types of antibody. Mention its function in immunity.

*6x4=24*

- D. Answer any 2 question from the following

5

1. What is the TCA cycle? Describe the Krebs cycle. Mention its importance. *1*  
 2. Describe the structure of DNA with a suitable diagram. What are the different forms of DNA? *(2+5+3)*  
 3. Mention the difference between Prokaryotic and Eukaryotic Cells. Write down the general features of Phylum Annelida. *(6+4)*  
*(5+5)*

10x2=20